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Highly Chemoselective Esterification from *O*-Aminoallylation of Carboxylic Acids: Metal- and Reagent-Free Hydrocarboxylation of Allenamides

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Abstract: Metal-free hydrocarboxylation of allenamides with various functionalized carboxylic acids were achieved with complete regio- and stereocontrol (>49:1). This environmentally compatible transformation affords γ-acyloxyenamides with exclusive *E*-selectivity. Electron rich, electron poor, aliphatic, aryl, and heterocyclic carboxylic acids all gave excellent yields (avg. 89%, 47 examples). We demonstrate the synthetic potential of this transformation in the late-stage modification of complex natural carboxylic acids and simple modification of the products to three-carbon synthons with ample opportunity for further diversification. DFT studies revealed that the reaction occurs in a stepwise manner through the intermediacy of a conjugated iminum species, which is rapidly captured by the carboxylate ion, resulting in the observed linear selectivity.

Introduction

The importance of ester groups has been highlighted by many applications in drug discovery,[1] polymerization,[2] and electronic material development.[3] Enamide groups have received particular attention as versatile ambiphilic building blocks[4] for biologically relevant molecules, including therapeutically important natural products and drug molecules.[5] Herein, we disclose a method for the direct *O*-aminoallylation of carboxylic acid groups using allenamides to afford highly functionalizable enamide derivatives.[6] This atom-economical method exhibits a broad substrate scope, exclusive chemo-, regio-, and *E*-selectivities. Our process transforms several inexpensive, stable, and abundant carboxylic acids into synthetically valuable γ-acyloxyenamides. The mild reaction conditions, exclusive chemoselectivity, and wide functional group tolerance also provide ready access to late-stage functionalization of complex natural products bearing carboxylic acids. This chemoselective functionalization is a powerful way to construct libraries of novel chemical entities from simple and complex carboxylic acid substrates,[7] enabling rapid exploration for structure–activity relationship (SAR).

The unique chemical features of allenes have inspired numerous atom- and step-economic strategies for constructing organic compounds including valuable synthons and polyfunctional structures.[8] Despite the great success of transition-metal catalysis, sustainable metal-free processes are increasingly in demand because of clear disadvantages of metals, such as: toxicity of metals, cost and poor availability of ligands, and hazardousness of additives. Meanwhile, Brønsted acid electrophilic activation of allenes as a greener alternative to transition metal functionalization are relatively rare.[9] In particular, hydrocarboxylation, which enables the synthesis of linear or branched allyl ester derivatives, under metal-free conditions remains unknown (Scheme 1 (A)). We report herein a greener alternative to the current state-of-the-art process, a metal-free hydrocarboxylation (Scheme 1 (B)), which could be used for efficient late-stage functionalization.

Transition-metal catalysts typically activate neutral (Scheme 1 (A), top). For example, iridium and rhodium[9b,c] catalyst processes that produce branched products have been reported independently by Krische[9a] and Breit.[9b,c] On the other hand, regiocontrolled synthesis of linear products with *trans* selectivity from allenes has been realized by Guo[9d] and Yamamoto,[9e,f] using silver and palladium catalysts, respectively. Most interestingly, literatures on transformations involving the allenamides are scarce. The one exception is the recent report of a copper-catalyzed hydrocarboxylation of allenamides discovered by the Monnier group,[9g] albeit with moderate yields, narrow substrate scope, and limited functional group compatibility (Scheme 1 (A), bottom).

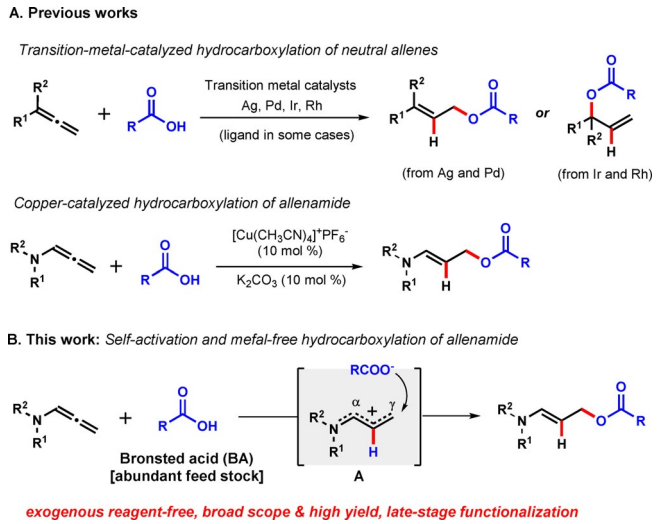
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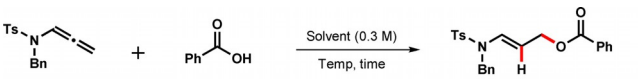
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Scheme 1. (A) Previous studies on allene hydrofunctionalization and (B) present reaction on hydrocarboxylation of allenamide.

Table 1. Optimization of metal-free hydrocarboxylation of allenamide **1a**.



allenamide (1a) benzoic acid (2a) γ-benzoyloxy (*E*)-enamide (3a)

Entry[a]	Solvent	*T* (°C)	Time (h)	Yield of **3a**[b] (%)
1	CH_2Cl_2	25	24	74[c]
2	CH_2Cl_2	55	12	81
3	**$(CH_2)_2Cl_2$**	**55**	**12**	**97 (96)**[d]
4	$CHCl_3$	55	12	93
5	toluene	55	16	71
6	THF	55	24	36[e]
7	1,4-dioxane	55	24	72[f]
8	DME	55	24	57[g]
9	**acetonitrile**	**55**	**24**	**96**
10	methanol	55	24	38
11	HFIP	55	1	trace[h]
12	*iso*-propanol	55	24	41
13	EtOAc	55	48	67[g]

[a] Unless otherwise noted, all reactions were performed using 0.1 mmol of **1a** and 0.13 mmol of **2a** in solvent (0.3 M). [b] Yield of **3a** was measured using 1,2-dibromomethane as an internal standard. [c] 23% of **1a** remained unreacted. [d] Isolated yield of **3a** after column chromatography. [e] 46% of **1a** remained unreacted and unidentifiable compounds formed. [f] 11% of **1a** remained unreacted. [g] 31% of **1a** remained unreacted. [h] Complete decomposition of **1a** was observed.

Our goal was to develop an orthogonal synthetic platform to these efforts. In contrast to neutral allenes, allenamides are known to be activated by a range of electrophiles, such as Lewis acids, Brønsted acids, and halonium reagents, to form α,β-unsaturated iminium intermediates, which can then be intercepted by various carbon and heteroatom (O or N) nucleophiles, including acids, alcohols, and amines.[10] We therefore envisioned activating the allenamide and trapping the resulting intermediate by a single reagent. Specifically, the carboxylic acid can protonate the allenamide to generate the reactive iminium intermediate (**A**, Scheme 1 (B)), which can then be intercept by the generated carboxylate ion.

Results and Discussion

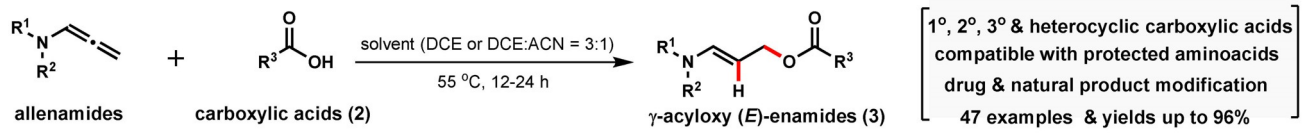
To develop the above proposed transformation, the reaction of allene sulfonamide **1a** with 1.3 equiv of benzoic acid (**2a**) in CH_2Cl_2 was first investigated (Table 1). To our delight, single regio- and stereoisomeric hydrocarboxylation product **3a** was obtained in 74% yield with 23% recovery of **1a** when the reaction mixture was stirred at room temperature for 24 h (entry 1). The reaction yield improved slightly to 81% with complete conversion of **1a** at 55 °C within 12 h (entry 2). The use of other chlorinated solvents improved the yield to as high as 97% (entries 3 and 4). Among the other tested solvents (entries 5–9), acetonitrile gave similar results, although with a longer incubation time (24 h). Changing to an alcohol solvent system did not improve the yield of hydrocarboxylation product **3a** (entries 10–12). Reaction with EtOAc was sluggish, with a 67% yield after two days (entry 13). We recognize that acetonitrile is environmentally more compatible than chlorinated solvents. In this work, we relied on DCE (entry 3) because of the faster reaction time.

With these optimized conditions in hand, the generality of the synthesis of *trans* γ-acyloxyenamides **3** was examined using various 1°, 2°, and 3° carboxylic acids **2** (Table 2). In general, all carboxylic acids reacted smoothly, regardless of the steric and electronic effects present. Specifically, aryl carboxylic acids with one or two electron-donating groups (Me, OMe), an electron-withdrawing group (Br, Cl, I, Ph, and NO_2), or both an electron-donating and electron-withdrawing group (Me and Br) positioned anywhere on the aryl ring were suitable substrates for this reaction (**3b–h**). Various phenylacetic acid derivatives were also effective and delivered the corresponding hydrocarboxylation products, **3i–k**, in similar yields (90–91%). The reaction was also compatible with aliphatic carboxylic acids irrespective of the nature or degree of α-substitution (**3l–o**). Similarly, unsaturated carboxylic acids featuring a proximal unsaturation (**3p**), a distal unsaturation (**3r**), and an unsaturation in a bridged carbocycle (**3q**) and a fatty acid (linoleic acid, **3s**) were all suitable coupling partners; no side reactions or geometric isomerization occurred. Bromo groups, α-hydroxyl groups at stereocenters, and phenolic groups were also compatible (**3t–v**). It is worth mentioning that the hydroxyl group of **2v** did not lower the chemoselectivity[11] preference for the acid, and the reaction proceeded rapidly to give the desired product within 30 min. These reaction conditions were fully compatible with mono-N-protected α-amino acids; benzoyl (on glycine), acetyl (on phenylalanine), and carbonate (on leucine and proline) protecting groups were all well tolerated under the standard conditions (**3w–z**). Most notably, the inertness of the halogen functionalities under these reaction conditions allows for significant facile post-functionalization using cross-coupling methodologies.

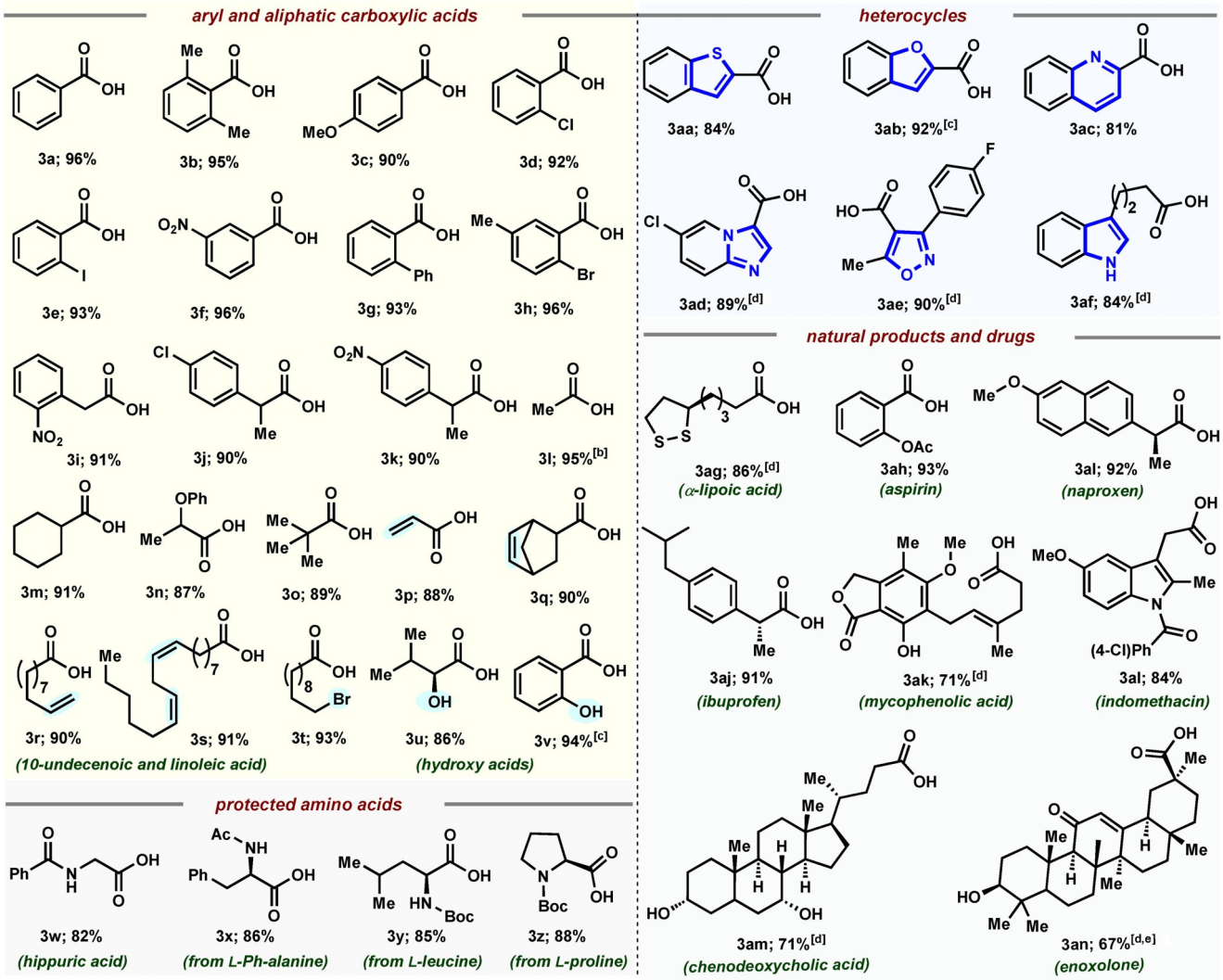
Next, we set out to explore the reactivity of heterocyclic carboxylic acids (Table 2 (A), top right). Hydrocarboxylation was efficient regardless of the electronic nature (O, S, or N), position,

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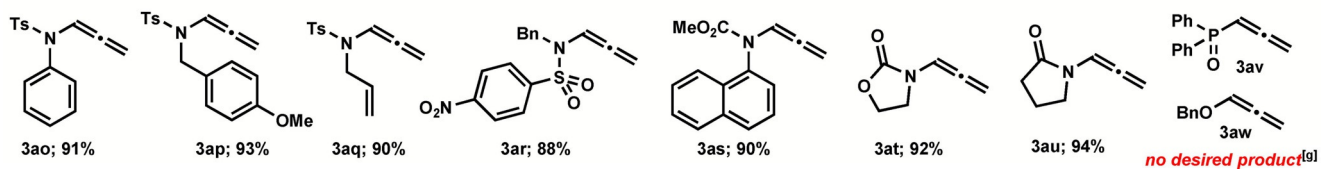
Table 2. Scope of (A) carboxylic acids and (B) allenamides.[a]



(A) Scope of carboxylic acids using allenamide 1a



(B) Scope and limitations of hydrocarboxylation using other allene derivatives[f]



[a] Experimental conditions: unless otherwise noted, all reactions were carried out under the optimal conditions using **1** (0.3 mmol) and **2** (0.39 mmol) in dichloroethane (DCE, 0.3 M). Isolated yields of the hydrocarboxylation products are given in brackets below each structure. [b] Yield was similar for 5 mmol scale reaction. [c] Reaction was complete within 30 min and the product was characterized without column chromatography. [d] Because of poor solubility, the reaction was carried out in a 3:1 DCE:acetonitrile (ACN) mixture. [e] Reaction was stirred for 24 h. [f] All reactions were carried out using benzoic acid **2a** (0.39 mmol) as the acid coupling partner. [g] While allenyl phosphine oxide remained intact, the OBn ether gave an unidentifiable mixture of products.

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and number of heteroatoms in the ring; all corresponding enamides (**3 aa**–**af**) formed as single stereo- (*E*) and regioisomers. Due to the remarkable functional group tolerance, exclusive selectivity, and promising yields, we were interested in late-stage modification of complex molecules rich in functional groups and complexity (Table 2 (A), bottom right). We therefore subjected several commercially available and widely prescribed drugs, α-lipoic acid (**2 ag**, antioxidant), aspirin, naproxen, ibuprofen, indomethacin (**2 ah**–**aj** and **2 al**, non-steroidal anti-inflammatory drugs), and mycophenolic acid (**2 ak**, immunosuppressant) to our standard conditions. Complex steroidal natural products chenodeoxycholic acid (**2 am**, bile acid) and enoxolone (**2 an**, antitussive) were also used. It is particularly noteworthy that enoxolone, a chiral tertiary carboxylic acid, also reacted efficiently and selectively. All substrates were transformed effectively into the enamide adducts with perfect chemo-, regio-, and stereoselectivity (**3 ag**–**an**); the various functionalities such as ethers, thioethers, esters, lactones, acetals, amides, olefins, and alcohols/phenols were inert under the reaction conditions, further demonstrating the synthetic utility of this transformation in complex scenarios. We next focused on expanding the generality of this transformation with other diverse allenamide derivatives (Table 2 (B)). As with the allene of *N*-benzyl toluenesulfonamide (**1 a**), allenamides bearing neutral or electron-rich aryls (**1 b** and **1 c**), allyl (**1 d**), nitrosulfonyl (**1 e**), acyclic or cyclic carbamates (**1 f** and **1 g**), or cyclic amide (**1 h**) groups were also suitable substrates and efficiently coupled with benzoic acid with similar reactivity and selectivity (**3 ao**–**au**). Unfortunately, allenyl phosphine oxide did not react to give **3 av**, while the allenyl ether **1 aw** gave a complex mixture.

We used density functional theory computations in order to elucidate the mechanism in this transformation. The structures, energies, and thermal corrections were computed at the PBE/6-31G* with SMD solvation corrections for DCE.[12] We computed the complete reaction coordinate diagram for **3 au** involving the oxazolidinone allenamide and benzoic acid. Four potential transition states ranging from concerted to stepwise mechanisms are shown in Figure 1. The most direct path to the experimentally observed γ-acyloxyenamides would be the concerted addition of the carboxylic acid O–H bond across the terminal alkene of the allenamide (**Concerted-1,2**). Alternatively, the addition could occur in a 1,4 fashion where the oxygen involved in the nucleophilic attack is distal to the OH involved in the protonation event (**Concerted-1,4**). A stepwise process is also possible (**Carboxylation First** vs. **Protonation first**). DFT computations revealed that the concerted process is not possible. The **Concerted-1,2** transition state was not a stationary point on the potential energy surface, and the **Concerted-1,4** transition state was geometrically impossible because the carboxylic acid group could not simultaneously reach the protonation and carboxylation sites. In the stepwise mechanisms, the **Carboxylation First** process did not lead to a stable intermediate—it monotonically fragmented back to the starting allenamide and carboxylic acid.

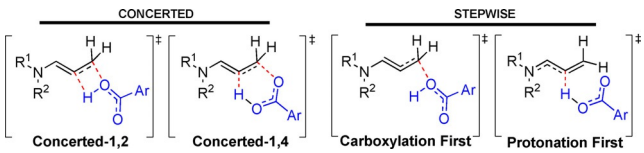


Figure 1. Possible transition state structures.

The only viable mechanism was the **Protonation First** pathway, where the carboxylic acid first activates the allenamide through protonation (Figure 2). The protonation of the allenamide had a barrier of 14.5 kcal mol^{-1} (**Cis-Prot-TS**), which led to the formation of the **Iminium-Carboxylate** complex ($\Delta G =$ 3.6 kcal mol^{-1}). This intermediate readily collapses to the product via 1,4-addition of the carboxylate to the vinylogous iminium terminus ($\Delta G = -20.2$ kcal mol^{-1}).

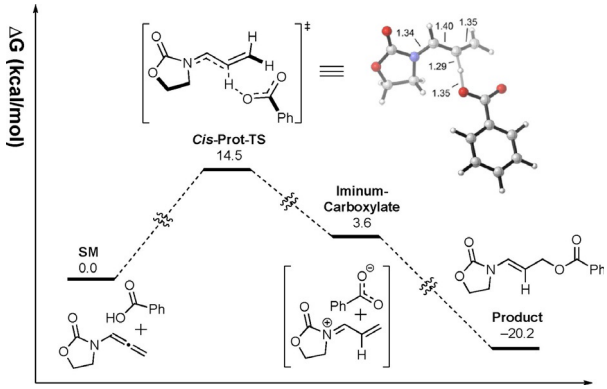


Figure 2. Computed reaction coordinate diagram and RDS-TS structure.

We juxtaposed our methodology to the work by Monnier (Table 3).[9g] Our methodology uniformly resulted in moderate to excellent yields (70–97 %). In comparison, the Cu-catalyzed hydrocarboxylation struggled with sterically hindered tertiary carboxylic acids, which were either completely unreactive (**3 o** and **3 an**) or led to moderate yields (**3 h**) (Table 3, entries 1, 2, and 7). In addition, α,β-unsaturated and indole-tethered acids (**3 p** and **3 af**, entries 3 and 4) were found to be partially incompatible with Cu-catalysis, as the former substrate polymerizes and the latter substrate deactivates the catalyst through N-chelation. Other sensitive functional groups incompatible with transition metal catalysts such as the cyclic disulfide of lipoic acid,[13] base-sensitive acyclic and cyclic esters, and hydroxy groups were not viable under the copper-catalyzed conditions; almost no product was formed in the case of **3 ag** and **3 ak** (entries 5 and 6). Lastly, highly functionalized tertiary carboxylic acid **3 an** did not afford the corresponding product under Cu catalysis (entry 7). In entries 8–11, we directly used the Monnier copper reaction substrates (**3 ao–au**) in order to directly compare the efficiencies of the reactions. In our hands, the metal-free protocol consistently gave higher yields.

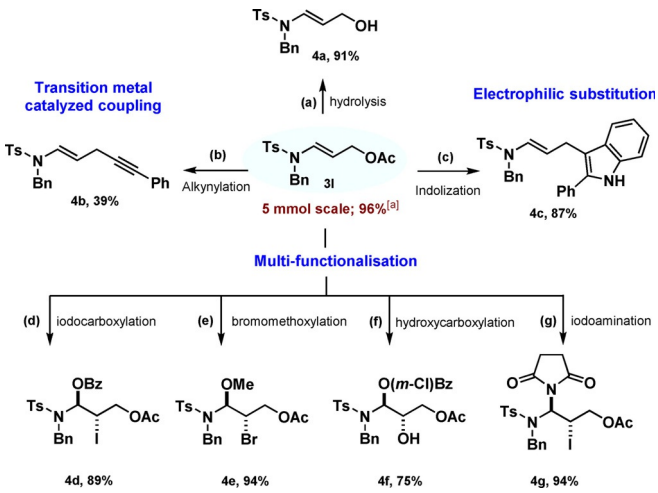
Finally, additional advantages of the present atom- and step-economical allenamide-carboxylic acid coupling process include scalability and product diversification (Scheme 2). A 5-mmol-scale reaction smoothly and cleanly afforded 1.7 g of **3 l** under the same conditions without the need for column chro-

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Table 3. Comparison of hydrocarboxylations of **1a** with acids **2** between our metal-free condition and previous Cu-catalyzed reaction condition.

Entry[a]	Product	^{1}H NMR yield (%)	
		Metal-free **(A)**	Cu-catalyzed **(B)**
1	**3h**	97	66
2	**3o**	90	0[b,c]
3	**3p**	92	32
4	**3af**	87	32
5	**3ag**	88	complex mixture
6	**3ak**	74	<5
7	**3an**	70	0[b,c]
8[d]	**3ao**	(91)	(43)
9[d]	**3aq**	(90)	(47)
10[d]	**3at**	(92)	(65)
11[d]	**3au**	(94)	(80)

[a] All reactions were performed using 0.1 mmol of **1a** under conditions **A** and **B** as described below. The conversion of **1** and yield of **3** were obtained via crude ^{1}H NMR analysis using an internal standard. **Condition A**: similar conditions to those mentioned in Table 2. **Condition B**: Cu-/base-catalyzed conditions reported in ref. 9g were employed. [b] As the reaction was sluggish, it was stirred for 36 h. [c] Even trace amounts of the desired product were not detected; NHTs(Bn) was formed in 91% yield for **3o** and 93% yield for **3an**. [d] Isolated yields of our condition and reported yields are shown in the parenthesis for direct comparison.



Scheme 2. Various transformations of **3l**. Unless otherwise noted, all reactions were performed using 0.2 mmol of **3l**. Reagents and conditions: (a) K_2CO_3 (3 equiv), MeOH, rt, 12 h. (b) Phenylacetylene (1.5 equiv), $ZnBr_2$ (20 mol%), diisopropylethylamine (1.5 equiv), $BF_3{\cdot}OEt_2$ (1.1 equiv), DCM, rt, 2 h. (c) 2-Phenylindole (1.1 equiv), $BF_3{\cdot}OEt_2$ (1.1 equiv), DCM, 0 °C, 5 min. (d) NIS (1.1 equiv), BzOH (1.3 equiv), DCM, 1 h, rt. (e) NBS (1.1 equiv), MeOH (0.3 M), 30 min, rt. (f) *m*-Chloroperbenzoic acid (1.3 equiv), DCM, 6 h, rt. (g) NIS (1.1 equiv), CH_3CN, 1 h, rt. [a] Gram-scale synthesis of **3l**: 5 mmol of **1a** was used under the standard conditions as described in entry 3 of Table 1.

matography purification. More importantly, diverse transformation to increase the structural and functional complexity of this synthetically versatile allyl acetate derivative was also feasible through nucleophilic and electrophilic transformations as well as transition-metal-catalyzed cross-couplings. Specifically, for compound **3l**, we first performed base-mediated deacylation, smoothly affording γ-hydroxyenamide **4a**, which can be manipulated further, in 91% yield. Lewis acid mediated S_N2-type functionalizations such as alkynylation[14] using a Zn-acetylide and arylation using an indole afforded ynenamide **4b** and enamide-tethered indoles **4c**, respectively. We also sought methods to construct modifiable three-carbon synthetic building blocks via electrophile-induced difunctionalization of the enamide unit under metal-free conditions. For example, electrophilic activation by N-halosuccinimides (NXS, X = I, Br) followed by their interception with oxygen nucleophiles (from a carboxylic acid or alcohol) allowed rapid construction of α-acetoxy-β-halo-*N*,*O*-acetals **4d** and **4e** in excellent yields (89% and 94%, respectively). Finally, atom-economic difunctionalization of **3l** to **4f** and **4g** through hydroxycarboxylation[15] and iodoamination, respectively, further demonstrated the synthetic utility of the hydrocarboxylation product.

Conclusions

We disclose a metal-free, mild, operationally simple hydrocarboxylation method with exclusive chemo-, regio-, and stereocontrol. DFT results show that the linear-selective hydrocarboxylation reaction proceeds in a stepwise manner through the intermediacy of a conjugated iminium species, which gives rise to a small interception barrier for the carboxylate ion to react. This atom-economical method can access a broad substrate scope, affording a wide variety of *trans*-γ-acyloxy-enamides in excellent yields. This coupling installs *trans*-propenamide unit in to carboxylic acid feedstocks proceeds with excellent functional group compatibility. Finally, the combined advantages of this methodology also allow ready access to late-stage functionalization of complex natural products bearing carboxylic acids, thereby enabling ample opportunity for facile downstream exploration and optimization of SAR.

Experimental Section

General Remarks: Unless otherwise mentioned, all reagents and solvents were purchased from commercial sources and were used without further purification. All reactions were performed using flame-dried glassware. The carboxylic acids were purchased from the commercial venders and used as such. Except for the gram scale reactions, all other reactions were performed in 7 mL screw capped vial. ^{1}H and ^{13}C NMR spectra were recorded at 25 °C on a 300 MHz NMR spectrometer in $CDCl_3$ and the residual solvent protons (7.26 for ^{1}H) or the solvent carbons (77.0 for ^{13}C) were used as internal references. Chemical shifts (δ) are reported in ppm. The coupling constants (*J*) are given in Hertz (Hz). The following abbreviations are used in the reporting NMR data: s, singlet; brs, broad singlet; d, doublet; t, triplet; q, quartet; qt, quintet; dd, doublet of doublets; dt, doublet of triplets; m, multiplet. High resolution mass spectra were recorded on a high-resolution Q-TOF mass spectrometer functioned by the electrospray ionization (ESI) technique. Melting points were reported in degree centigrade (°C) within ±2 range. The details of experimental procedures and characterization data (^{1}H, ^{13}C ESI-HRMS and m.p, if solid) of all the new compounds were in the Supporting Information.

Representative procedure for hydrocarboxylation: To a screw capped 7 mL vial, allenamides **1** (0.3 mmol) and carboxylic acids **2** (0.39 mmol) were dissolved in 1 mL (0.3 M) of dichloroethane (DCE). The above solution was continued to stir for 12 h at 55 °C

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(with few exception, see Supporting Information). When the consumption of all the starting allenamide was observed (monitored by TLC), the reaction mixture was diluted with 3 mL of DCE followed by addition of saturated $NaHCO_3$ solution (2 mL). After vigorous stirring for one more hour, the organic solvent was separated. The remaining aqueous layer was further extracted with DCM (2 X 5 mL). The solvent was removed from the collected organic layer to give a pale yellow liquid, which on silica gel column chromatography furnished the desired γ-carboxylation products up to 96% yield.

Gram scale synthesis of hydrocarboxylation product 3l: To a 25 mL round bottom flask containing allenamide **1a** (1.495 g, 5 mmol) in dichloroethane (0.4 M, 16.7 mL) was added acetic acid **2l** (0.37 mL, 6.5 mmol) at room temperature. The above reaction mixture was heated at 55 °C for 12 h. It was then cooled to room temperature prior to the addition of saturated aqueous $NaHCO_3$ solution (15 mL). To quench the unreacted acetic acid stirring was continued for one more hour. The organic layer was separated and the aqueous layer was further extracted with DCM (2×20 mL). The combined organic layer was concentrated to remove all the solvent and dried under vacuum to give 1.7 g of hydrocarboxylation product **3l** as an off-white solid (95% yield). The purity of the analytical data is similar to that obtained from column chromatography purification. This compound was then directly used for different synthetic transformations.

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Conflict of interest

The authors declare no conflict of interest.

Keywords: Brønsted acid · carboxylic acids · (*E*)-enamides · linear selective · metal-free

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